12. 9/3/26

① The sum of the present age of a father and his son is 60 years. Six years ago, the father's age was 5 times age of the son. After 7 years the son age will be?

Father	Son
x	y
$60-y$	y
	$y-6$
	$y-6$

$x+y=60$
$x=60-y$

6 years ago, $60-y-6$

$54-y$

$54-y = 5(y-6)$

$54-y = 5y-30$

$54+30 = 5y+y$

$84 = 6y$

$y = 84/6 = 14$

7 years after $= 14+7$
$= 21.$

② 10 years ago, A was half on B in age. If the ratio of their present ages is 3:4, what will be to total of their present ages?

	A	B
present ages	$3x$	$4x$
10 years ago	$3x-10$	$4x-10$

$3x-10 = \frac{1}{2}(4x-10)$

$3x-10 = 2x-5$

$3x-2x = -5+10$

$\boxed{x = 5}$

A => $3\times5 = 15$

B => $4\times5 = 20.$

③ By selling bouquets for Rs. 63 a florist agains 5% at what price should he sell the bouquets to gain 10% on the cp?

Formula :-
cp => find
$s.p = c.p \times \left(1 + \frac{profit}{100}\right)$

selling price : 63
cost price : ?
profit = 5%/100

$63 = cp \times \left(1 + \frac{5}{100}\right)$

$63 = cp \times 1.05$

$\boxed{cp = \frac{63}{1.05} = 60}$

$\boxed{63 = 60 \times 1.05}$

$s.p = 60 \times 1.10$

$\boxed{s.p = 66}$

④ in a maternity center, 5% of all childbirth cases result in twins. what is the approximate percent of twins out of the total children born.

twins → 5%
take 100 childbirth → 100
↳ 100−5 = 95
single children → 95
twins children : 5×2 = 10
total children → 95+10 = 105
find percent of twins :-

$\frac{10}{105} \times 100 = 20$
21

= 9.5%

⑤ In a mixture of petrol and kerosene petrol is only 99 litres. If this same quantity of petrol would be present in another mixture whose total volume is 198 litres less than the actual mixture, then the ~~actual~~ concentration of petrol in the actual mixture would have been 13.33 percentage points less than that of the new mixture. what is the concentration of petrol in the actual mixture?

⑥ The sum of ages J and her mother is 36 years, also 6 years ago, the mother's age was 3 times Jenny's age. find the present age of J's mother?

J	M
~~y~~	~~J-36~~
x	y
y-36	y
y-30	y-6

$3(y-30) = y-6$

$3y-90 = y-6$

$2y = -6+90$

$2y = 84$

$y = 42$

x y

$J+M = 36$

$m = J-36$

$y = x-36$

$x+y = 36$

$x = y-36$

J	M
36-m	m
30-m	m-6

$90-3m = m-6$

$96 = 4m$

$\boxed{m = 24}$

⑦ Kevin Rudd is older than John howard by 8 years. If their ages are in the ratio 6:5 - how old is Kevin Rudd, 6 years hence.

Ratio : 6:5

$6x \rightarrow$ Kevin

$5x \rightarrow$ John

age diff :

$6x - 5x = 8$

$\boxed{x = 8}$

present age :- $6 \times 8 = 48$

$5 \times 8 = 40$

after 6 years, $48 + 6 = 54$

8) In an election, candidate A got 40% of the total valid votes. If 55% of the total votes were declared invalid and the total no. of votes is 280000, find the no. of valid votes polled in favour of the candidate.

total valid votes $= 40/100$

55% invalid so valid votes $= 45/100$

no. of votes $= 280,000$

$= \frac{40}{100} \times \frac{45}{100} \times 280000$

$= 50,400$.

9) The present age of a father is 3 years more than three times the age of his son. Three years hence, father's age will be 10 years more than twice the age of the son. find the present age of the father?

Father	Son
$3x+3$	x

after 3,

$3x+6$ $\quad$ $x+3$

$3x+6 = 2(x+3)+10$

$3x+6 = 2x+6+10$

$3x-2x = 16-6$

$1x = 10$

$\boxed{x = 10}$

$\boxed{3(30)+3 = 33 \text{ years.}}$

10) The price of wheat increases by 40%. later the price falls to the original price. By what percentage should the new price fall?

assume original price = 100

increase = 40%

new price = 140

$\frac{40}{140} \times 100$

$= 28.57\%$